

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12 ASSESSMENT TASK 3

JUNE 2003

MATHEMATICS

Extension 1

Time Allowed: 70 minutes

Instructions:

- Attempt all questions
- Start each question on a new page
- Show all necessary working
- Marks may be deducted for careless or badly arranged work
- Approved calculators may be used
- Standard integrals are attached and may be removed for your convenience.

Name: _____

Teacher: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total

Question 1

- a) Consider the function $f(x) = 2 \tan^{-1} x$.
- (i) Evaluate $f(\sqrt{3})$ (1)
 - (ii) Draw the graph of $y = f(x)$, labelling any key features (2)
 - (iii) Find the slope of the curve at the point where it cuts the y axis. (2)
- b) (i) $\int \frac{dy}{\sqrt{16 - y^2}}$ (1)
- (ii) $\int \frac{dt}{5 + t^2}$ (2)
- c) By letting $A = \cos^{-1}(\frac{3}{7})$, find the exact value of $\sin A$. (2)

Question 2

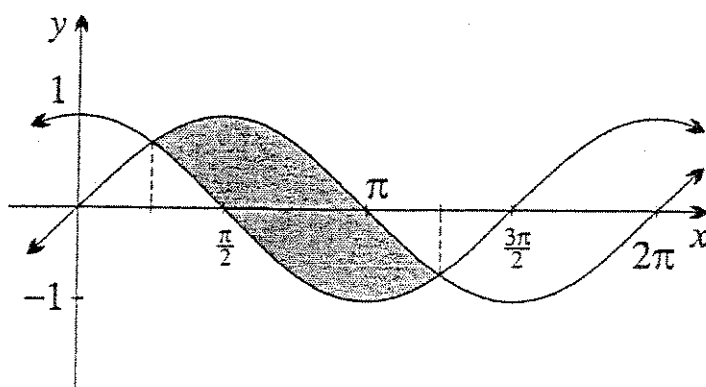
- a) Differentiate $e^{2x} \ln x$ (2)
- b) Evaluate $\int_0^1 \frac{2x}{(2x+1)^2} dx$ by using the substitution $v = 2x+1$ (4)
- c) Solve for x in exact form: $e^{2x} - 3e^x + 2 = 0$ (3)

Question 3

- a) The polynomial $P(x) = x^4 - 3x^3 + ax^2 + bx - 6$ leaves a remainder of 8 when divided by $x + 1$. If $x - 3$ is a factor for $P(x)$, find the values of a and b . (3)
- b) Two of the roots of the equation $x^3 + ax^2 + b = 0$ are reciprocals of each other (a, b both real).
- Show that the third root is equal to $-b$. (1)
 - Show that $a = b - \frac{1}{b}$. (2)
 - Show that the two roots, which are reciprocals, will be real if $-\frac{1}{2} \leq b \leq \frac{1}{2}$. (3)

Question 4

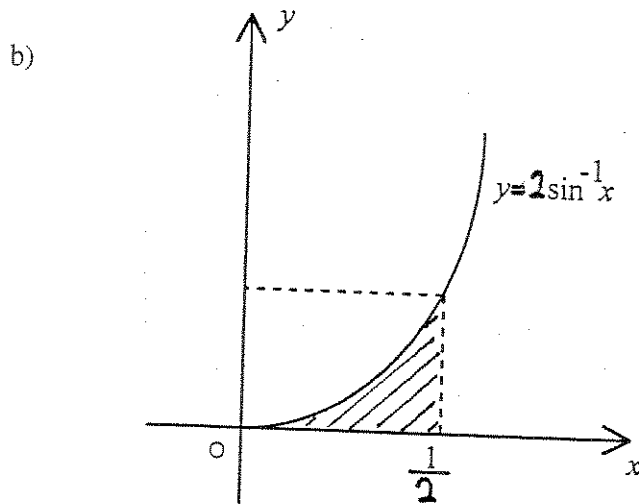
- a) Write a result for $\sin^2 3x$ in terms of $\cos 6x$. (1)
- Hence evaluate $\int_0^{\frac{\pi}{2}} \sin^2 3x dx$ (2)
- b) Find the area of the shaded region below: (4)



- c) Solve $\log_e(x+1) + \log_e x = \log_e 2$ (3)

Question 5

- a) (i) Find the domain of $y = \sin(\sin^{-1} x)$ (1)
 (ii) Hence sketch the function (1)



The sketch shows the graph of the curve $y = f(x)$ where $f(x) = 2 \sin^{-1} x$.

The area under the curve for $0 < x < \frac{1}{2}$ is shaded.

- (i) Find $f(\frac{1}{2})$ (1)
 (ii) Find $f^{-1}(x)$ (1)
 (iii) Hence or otherwise, calculate the area of the shaded region. (3)
- c) Give the general solution to the equation $\tan \vartheta = 1$ (ϑ in radians) (2)

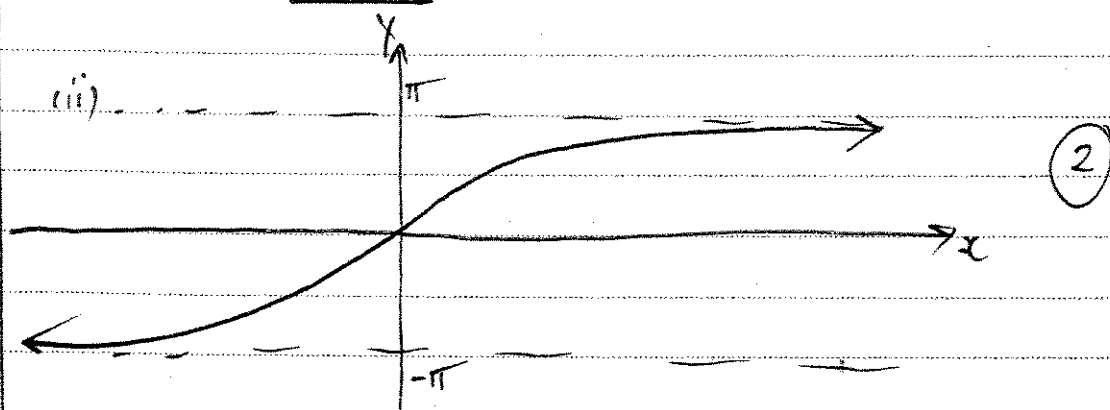
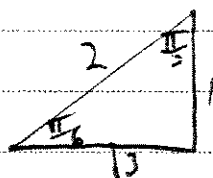
Question 6

- a) (i) Differentiate $y = \cot 2x$ (2)
 (ii) Hence or otherwise find $\int \operatorname{cosec}^2 2x \, dx$ (2)
- b) Consider the function $y = \cos x - \frac{1}{4\sqrt{3} \sin x}$
- (i) Verify that $\frac{dy}{dx} = 0$ when $x = \frac{\pi}{6}$ (3)
 (ii) Sketch the curve over the domain (3)
 $0 < x \leq \frac{\pi}{2}$, given that $\frac{d^2y}{dx^2} < 0$.

On your sketch, write the coordinates of the turning point in exact form and label the asymptote (the x intercepts are not required).

Solutions

Q1 (a) (i) $f(\sqrt{3}) = 2 + \tan^{-1} \sqrt{3}$
 $= 2 \times \frac{\pi}{3}$
 $= \frac{2\pi}{3}$ (1)

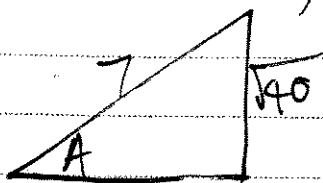


(iii) $f'(x) = 2 \times \frac{1}{1+x^2}$ (1)
 $f'(0) = 2$ when $x=0$. (1)

(b) (i) $\int \frac{dy}{\sqrt{16-y^2}}$ (1)
 $\sin^{-1}\left(\frac{y}{4}\right) + C$

(ii) $\int \frac{dt}{5+t^2}$ (2)
 $= \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{t}{\sqrt{5}}\right) + C$

(c) $A = \cos^{-1}\left(\frac{3}{7}\right)$



$\sin A = \frac{\sqrt{40}}{7}$ (2)

Q2. (a) (i) $\frac{d}{dx}(e^{2x} \ln x)$
 $uv' + v u'$
 $= e^{2x} \times \frac{1}{x} + \ln x \times 2e^{2x}$
 $= \frac{e^{2x}}{x} + 2e^{2x} \ln x$ (2)

$$(b) \int_0^1 \frac{2x}{(2x+1)^2} dx$$

$$\text{Let } u = 2x+1$$

$$du = 2 dx$$

$$\frac{dx}{du} = \frac{du}{2}$$

$$\int \frac{u-1}{u^2} \times \frac{du}{2} \quad (1)$$

$$\frac{1}{2} \int \frac{1}{u} - \frac{1}{u^2} du$$

$$\frac{1}{2} \left[\log_e u + \frac{1}{u} \right] \quad (1)$$

$$\frac{1}{2} \left[\log_e(2x+1) + \frac{1}{2x+1} \right]_0^1 \quad (1)$$

$$\frac{1}{2} \left[\log 3 + \frac{1}{3} - (\log 1 + 1) \right]$$

$$\frac{1}{2} \left[\log 3 - \frac{2}{3} \right] \quad (1)$$

$$(c) e^{2x} - 3e^x + 2 = 0$$

$$(e^x)^2 - 3e^x + 2 = 0$$

$$u^2 - 3u + 2 = 0 \quad (1)$$

$$(u-1)(u-2) = 0$$

$$e^x = 1, e^x = 2 \quad (1)$$

$$\therefore x = 0 \text{ or } \ln 2 \quad (1)$$

$$(Q) 3 (a) P(-1) = 8 = (-1)^4 - 3(-1)^3 + a(-1)^2 + b(-1) - 6$$

$$8 = 1 + 3 + a - b - 6$$

$$10 = a - b \quad (1)$$

$$P(3) = 0 = 81 - 81 + 9a + 3b - 6$$

$$2 = 3a + b \quad (2)$$

(1) for 2 equations each

$$(1) + (2) \Rightarrow 12 = 4a$$

$$a = 3 \therefore b = -7 \quad (1)$$

$$(b) x^3 + ax^2 + b = 0$$

Let roots be

$$\alpha, \frac{1}{\alpha}, \beta$$

(i) product of roots equals $-\frac{d}{a} = -b$

$$\therefore \beta = -b \quad (1)$$

(ii) Sum of roots:

$$\frac{1}{\alpha} + \alpha + \beta = -a \quad (1)$$

Sum of roots (pairs)

$$1 - \alpha\beta - \frac{b}{a} = 0$$

$$(1) \quad 1 - b(\alpha + \frac{1}{\alpha}) = 0 \quad (2)$$

(1) into (2)

$$1 - b(b - a) = 0$$

$$1 - b^2 + ab = 0$$

$$\begin{aligned} ab &= b^2 - 1 \\ a &= b - \frac{1}{b} \end{aligned} \quad \begin{aligned} &\text{as eqn (1)} \\ &\text{as eqn (2)} \end{aligned}$$

$$\text{iii) } x^3 + ax^2 + b = 0$$

$-b$ is a root of the above

$\therefore (x+b)$ is a factor

$$\therefore \text{Let } x^3 + ax^2 + b = (x+b)(x^2 + Px + Q) \quad (1)$$

Equating coefficients:

$$\Rightarrow b = bQ \quad \text{also } 0 = Q + Pb$$

$$\therefore Q = 1$$

$$\therefore P = -\frac{Q}{b} = -\frac{1}{b}$$

Hence $x, \frac{1}{x}$ are roots of the quadratic

$$x^2 - \frac{1}{b}x + 1 = 0$$

$$\text{Here, } \Delta = \frac{1}{b^2} - 4$$

Real roots if $\Delta \geq 0 \quad (1)$

$$\frac{1}{b^2} - 4 \geq 0$$

$$\frac{1}{b^2} \geq 4, \quad b^2 \leq \frac{1}{4} \Rightarrow \underline{\underline{-\frac{1}{2} \leq b \leq \frac{1}{2}}} \quad (1)$$

$$\text{Q 4a) } \cos 2x = 1 - 2\sin^2 x$$

$$\therefore \cos 6x = 1 - 2\sin^2 3x$$

$$\therefore \sin^2 3x = \frac{1 - \cos 6x}{2} \quad (1)$$

$$\therefore \int_0^{\frac{\pi}{2}} \sin^2 3x dx \text{ becomes } \int_0^{\frac{\pi}{2}} \frac{1 - \cos 6x}{2} dx$$

$$= \left[\frac{x}{2} - \frac{\sin 6x}{12} \right]_0^{\frac{\pi}{2}} \quad (1)$$

$$= \frac{\pi}{4} - \frac{\sin 3\pi}{12} - \left(0 - \frac{\sin 0}{12} \right)$$

$$= \underline{\underline{\frac{\pi}{4}}} \quad (1)$$

$$(b) A = \int_a^b \sin x - \cos x \, dx$$

a, b from $\sin x = \cos x$

$$\tan x = 1$$

$$\underline{x = \frac{\pi}{4}, \frac{5\pi}{4}} \quad (1)$$

$$A = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \sin x - \cos x \, dx \quad (1)$$

$$\left[-\cos x - \sin x \right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \quad (1)$$

$$= -\cos \frac{5\pi}{4} - \sin \frac{5\pi}{4} - \left(-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \right)$$

$$= -\left(-\frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right)$$

$$= \underline{2\sqrt{2} \text{ units}^2} \quad (1)$$

$$(c) \log_e(x+1) + \log_e x = \log_e 2$$

$$\log_e [x(x+1)] = \log_e 2$$

$$x^2 + x = 2 \quad (1)$$

$$x^2 + x - 2 = 0$$

$$(x-1)(x+2) = 0 \quad (1)$$

$$x = 1 \text{ or } -2 \text{ as } x \text{ must be } > 0$$

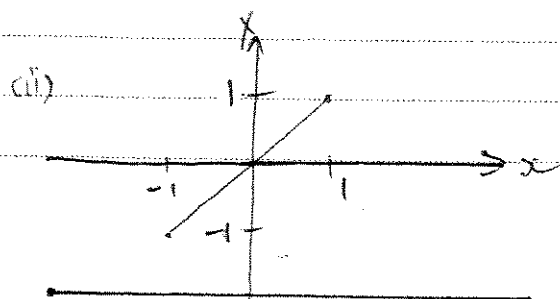
$$\therefore \underline{x = 1} \quad (1)$$

$$Q 5(a) y = \sin(\sin^{-1} x)$$

$$-1 < x < 1 \text{ for } y = \sin^{-1} x$$

$$-1 < \sin x < 1 \text{ for } y = \sin^{-1} x$$

$$\therefore \underline{-\frac{\pi}{2} < x < \frac{\pi}{2}} \quad (1)$$



(1)

$$\textcircled{b} \text{ (i) } f\left(\frac{1}{2}\right) = 2 \sin^{-1} \frac{1}{2} \\ = 2 \times \frac{\pi}{6} \\ = \frac{\pi}{3} \quad \textcircled{1}$$

$$\text{ (ii) } y = 2 \sin^{-1} x \\ x = 2 \sin^{-1} y \\ \frac{x}{2} = \sin^{-1} y \\ \sin \frac{x}{2} = y \quad \textcircled{1}$$

$$\text{ (iii) } A = \text{Rectangle} - \int x \, dy \\ = \frac{1}{2} \times \frac{\pi}{3} - \int_0^{\frac{\pi}{3}} \sin \frac{y}{2} \, dy \quad \textcircled{1} \\ = \frac{\pi}{6} - \left[-2 \cos \frac{y}{2} \right]_0^{\frac{\pi}{3}} \quad \textcircled{1} \\ = \frac{\pi}{6} + \left[2 \cos \frac{y}{2} \right]_0^{\frac{\pi}{3}} \\ = \frac{\pi}{6} + \left[2 \cos \frac{\pi}{6} - 2 \cos 0 \right] \\ = \frac{\pi}{6} + \sqrt{3} - 2 \quad \textcircled{1}$$

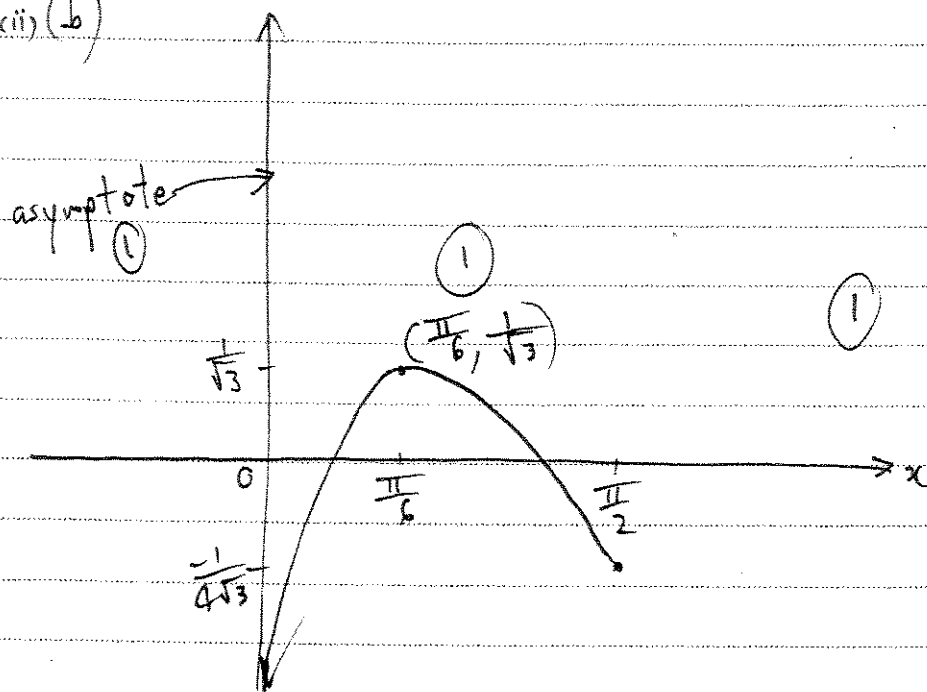
$$\text{ (c) } \tan \theta = 1 \\ \theta = \tan^{-1}(1) + n\pi \quad \textcircled{1} \\ = \frac{\pi}{4} + n\pi \quad \text{where } n = 0, 1, 2, 3$$

$$\textcircled{Q6} \text{ (i) } y = \cos x - \frac{1}{4\sqrt{3} \sin x}$$

$$\frac{dy}{dx} = -\sin x + \frac{\cos x}{4\sqrt{3}(\sin x)^2} \quad \textcircled{2}$$

$$\text{when } x = \frac{\pi}{6} \\ \frac{dy}{dx} = -\sin \frac{\pi}{6} + \frac{\cos \frac{\pi}{6}}{4\sqrt{3}(\sin \frac{\pi}{6})^2} \\ = -\frac{1}{2} + \frac{\frac{\sqrt{3}}{2}}{4\sqrt{3} \times \frac{1}{4}} \\ = -\frac{1}{2} + \frac{1}{2} = 0 \quad \textcircled{1}$$

(ii) (b)



① for slope

$$\begin{aligned}
 \text{When } x = \frac{\pi}{6}, \quad y &= \cos \frac{\pi}{6} - \frac{1}{4\sqrt{3} \sin \frac{\pi}{6}} \\
 &= \frac{\sqrt{3}}{2} - \frac{1}{4\sqrt{3} \times \frac{1}{2}} \\
 &= \frac{\sqrt{3}}{2} - \frac{1}{2\sqrt{3}} \\
 &= \frac{6 - 2}{4\sqrt{3}} \\
 &= \frac{1}{\sqrt{3}}
 \end{aligned}$$

Q6

(i) $y = \cot 2x$

$$\frac{dy}{dx} = -\operatorname{cosec}^2 2x \times \frac{d}{dx}(2x)$$

$$\frac{dy}{dx} = -2 \operatorname{cosec}^2 2x \quad (2)$$

(ii) $\therefore -\frac{1}{2} \frac{d}{dx}(\cot 2x) = \operatorname{cosec}^2 2x \quad (1)$

$$\therefore \int -\frac{1}{2} \frac{d}{dx}(\cot 2x) = \int \operatorname{cosec}^2 2x \, dx$$

$$\underline{-\frac{1}{2} \cot 2x + c} = \int \operatorname{cosec}^2 2x \, dx.$$

①